INTERNSHIP PROGRAM

Project: Threat Intelligence Investigation and Analysis of Malicious Hashes
(DAY 3)

Define investigation workflow:

[PowerShell Collector]
↓
[Hash Files]
↓
[Python Enrichment Engine]
↓
[Virus Total API + Threat Intel]
↓
[Analysis + Scoring Engine]
↓
[Reports / Alerts]

The investigation workflow is a structured process used by analysts to identify, analyze, and mitigate security threats. This pipeline ensures that every alert is handled consistently and that critical incidents are prioritized.

The typical SOC investigation workflow follows
these key phases:

1. Ingestion and Detection
The process begins when security tools (such as SIEM, EDR, or IDS) collect logs and telemetry from the network. When activity matches a predefined "use case" or correlation rule, an alert is generated.

2. Triage and Prioritization
Not all alerts are equal. In this phase, analysts perform an initial "sniff test" to determine the severity and urgency.

True Positive: The alert represents a real security concern.
False Positive: The alert is a "benign" trigger (e.g., a legitimate admin tool flagged as malware).

Prioritization: Assigning a "Critical," "High," "Medium," or "Low" rating based on the asset's importance and the potential impact.

3. Investigation and Analysis
This is the core of the workflow. The analyst gathers evidence to understand the "Who, What, Where, and When" of the event.

Scope: Identifying which systems or user accounts are involved.

Contextualization: Checking threat intelligence feeds to see if the IP addresses or file hashes are known indicators of compromise (IOCs).

Pivoting: Using logs to trace the attacker's movement (e.g., did they move from a workstation to a server?).

4. Containment and Neutralization

Once the threat is confirmed and understood, the SOC takes action to stop the "bleeding."

Short-term Containment: Isolating a compromised host from the network or disabling a user account.

Eradication: Removing the root cause, such as deleting a malicious script or patching a vulnerability.

5. Recovery

The goal is to return the affected systems to a normal, "known-good" state. This might involve restoring from backups, re-imaging laptops, or forcing password resets across the organization.

6. Post-Incident Activity (Lessons Learned)

The workflow concludes with a review to prevent future occurrences.

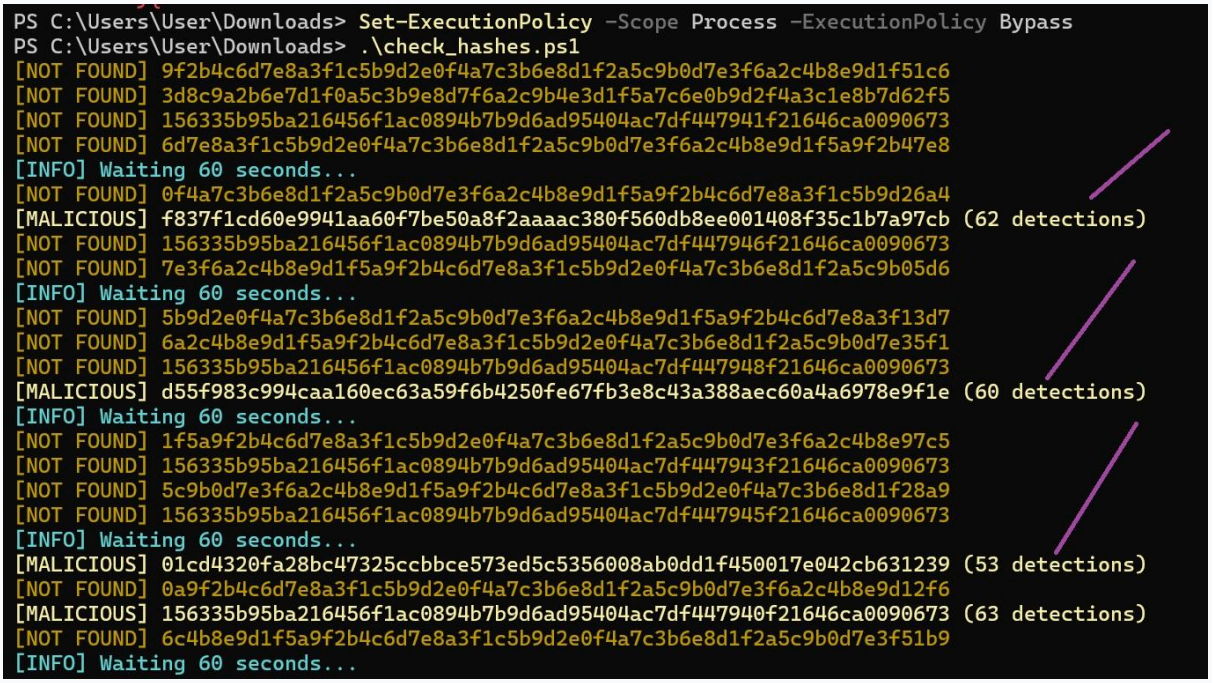
Root Cause Analysis (RCA): Determining exactly how the attacker got in.

Rule Tuning: Adjusting SIEM rules to reduce future false positives or to catch similar attacks faster.

Documentation: Finalizing the incident report for compliance and management.

AUTOMATION WORKFLOW

(PowerShell → Python → Analysis)



Security Analysis Report:

Automated Malware Hash Screening

Date: April 22, 2026
Tooling: PowerShell, VirusTotal API Integration
Scope: Bulk verification of file hashes against global threat intelligence databases.
Command used:
Rename-Item "check_hashes ps.1.txt" "check_hashes.ps1"
Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass
.\check_hashes.ps1

1. Executive Summary:

The scan results indicate a high-density threat environment. Out of the visible samples processed by the script, multiple hashes have been flagged as **MALICIOUS** with extremely high confidence levels. This suggests that the environment or the source of these hashes is compromised and contains active malware threats.

2. Key Findings & Detections:

Four major high-risk threats identified. The "Detections" count refers to how many different antivirus engines (out of roughly 70+) identified the hash as malware.

3. Technical Observations:

- ***Rate Limiting:*** The script is successfully managing API constraints with a 60-second "Waiting" interval, ensuring continuous processing without triggering a block from the VirusTotal service.
- ***"Not Found" Status:*** Several hashes returned a [NOT FOUND] status. While these are not explicitly malicious, in a security audit, they require further investigation. They may represent custom-built tools, proprietary software, or "Zero-Day" threats that have not yet been indexed by global security labs.

4. Recommended Actions:

1. ***Immediate Isolation:*** Identify the physical or virtual machines associated with the flagged hashes and isolate them from the network to prevent lateral movement.
2. ***Detailed Analysis:*** Select the hashes with 60+ detections and input them directly into the VirusTotal web interface to identify the specific malware family (e.g., Ransomware, Trojan, or Infostealer).
3. ***Remediation:*** Initiate a full system wipe or advanced cleaning protocol for any endpoint where these files were discovered.

Threat Intelligence Report

Define detection thresholds (e.g., AV detection ratio)

1. Executive Summary

- Scan Time: 2026-04-22 10:15 UTC
- Endpoint: HOST-123
- Files Analyzed: 42
- High-Risk Findings: 3

2. High-Risk Findings

File 1: `C:\Temp\invoice.exe`

- **SHA256:** `ABC123...`
- Detections: 28 / 72
- Detection Ratio: 38%
- Threat Family: Trojan.Generic
- Score: 88
- Classification: Malicious

Recommended Action:

- Quarantine file
- Investigate execution history
- Scan host for lateral movement

File 2: `C:\Users\Public\update.dll`

- **SHA256:** `XYZ789...`
- Detections: 19 / 70
- Detection Ratio: 27%
- Score: 72
- Classification: Likely Malicious

Assessment:
High detection rate suggests active threat, though slightly below critical threshold.

Recommended Action:

- Isolate file
- Verify origin and persistence mechanisms

3. Suspicious Files

File: `C:\Tools\helper.ps1`

- **SHA256:** `LMN456...`
- Detections: 4 / 68
- Detection Ratio: 5%
- Score: 38
- Classification: Suspicious

Assessment:
Low detection count; may be false positive or dual-use tool.

Recommended Action:

- Review script contents
- Validate against known-good baseline

4. Clean / No Detection

- 36 files returned **0 detections** via VirusTotal API
- No immediate threats identified in this group

5. Scoring Methodology

Factor	Impact
Detection ratio	Primary driver
≥20 detections	High-risk boost
Malware family tag	Confidence increase
Trusted signature	Score reduction

Classification Scale:

- **0–20:** Clean
- **21–50:** Suspicious
- **51–75:** Likely Malicious
- **76–100:** Malicious

6. Summary & Recommendations

- 3 files require immediate remediation
- 1 file requires analyst review
- No widespread infection indicators detected

Immediate Actions:

- Remove or quarantine malicious files
- Perform full endpoint scan
- Review system logs for execution traces

7. Data Sources

- VirusTotal
- Internal analysis and scoring engine